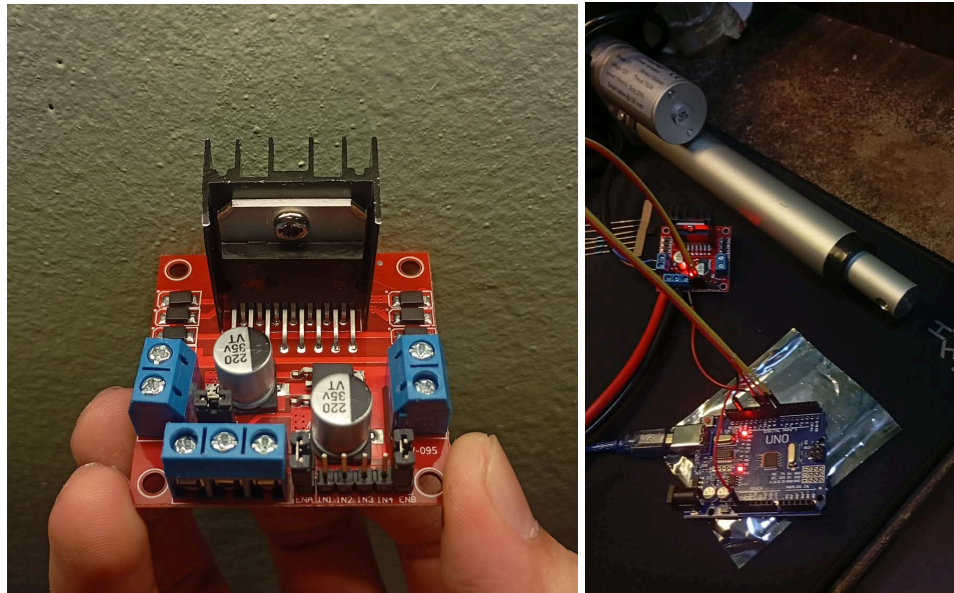


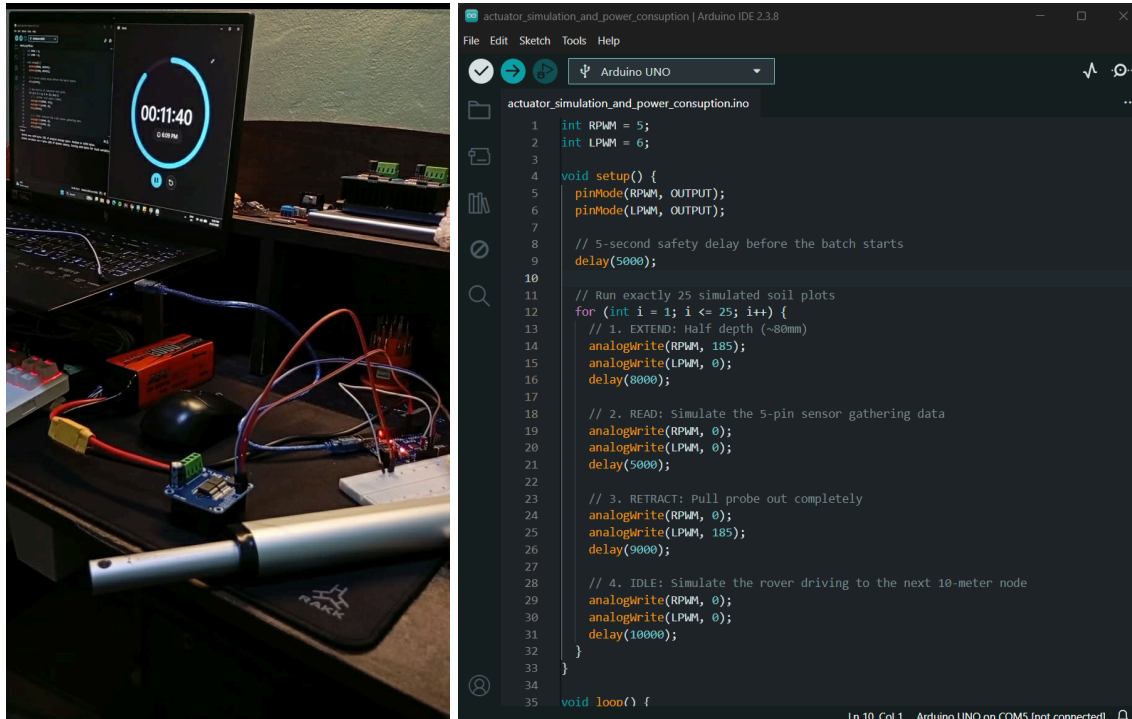
Linear Actuator Testing

Phase 1: Initial Testing & Hardware Failure Incident (L298N)



- **Hardware Setup:** Arduino Uno, 12V Hawiko Linear Actuator (700N, 150mm stroke), L298N Motor Driver (2A max), and a 14.8V 6000mAh 70C LiPo Battery.
- **Incident:** During manual push-button testing, the actuator experienced a transient stall. To isolate the issue, a direct DC polarity test was conducted using the L298N terminal blocks as a touch-point. Immediately upon connecting the LiPo battery, the L298N motor driver suffered catastrophic thermal failure and emitted smoke.
- **Root Cause:** The failure was caused by a severe current mismatch. **A stalled 700N actuator draws a massive current spike.** The high-discharge 70C LiPo supplied the requested current instantly, surpassing the L298N's strict 2A limit and destroying the H-bridge integrated circuit. The LiPo battery and actuator were unharmed.
- **Corrective Action:** The L298N was replaced with a **Double BTS7960 43A** High-Power Motor Driver. To prevent overvolting the 12V actuator with the 16.8V (peak) LiPo, the software was updated with a PWM value of 185 (roughly 72% duty cycle) to safely scale the output to exactly 12V.

Phase 2: Power Consumption & Endurance Simulation (BTS7960)



Methodology

- A batch simulation script was uploaded to the Arduino. The code **executed 25 continuous sampling cycles mimicking the 10-meter node intervals**.
- **8-second extension (half-stroke to ~80mm)**
- **5-second pause to simulate the multi-metric soil sensor gathering data,**
- **9-second full retraction,**
- **a 10-second idle period.**
- Power consumption was measured using a digital capacity checker on a single 14.8V 6000mAh LiPo battery.

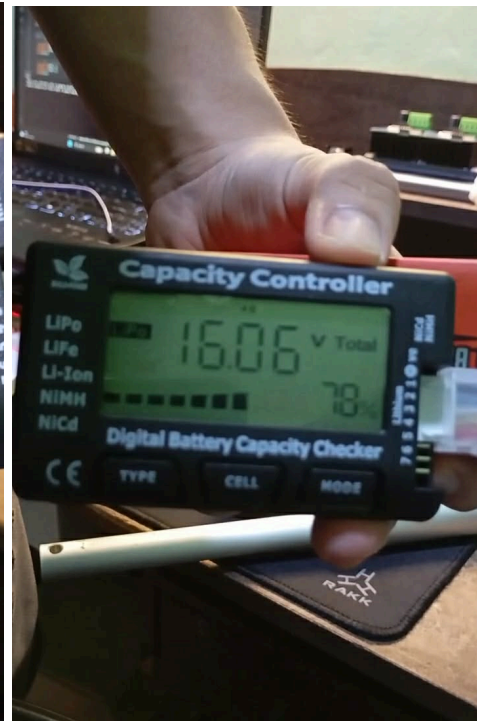
Observations: The BTS7960 large aluminum heatsink remained completely cool throughout the 13.5-minute test, validating the thermal stability of the upgraded driver.

Results:

- Starting Battery State: 16.10V / 79% capacity.
- Ending Battery State (**after 2 minutes of resting**): 16.06V / 78% capacity.
- Total Capacity Consumed: 1% (60mAh) **over 25 cycles**.
- Calculated Consumption Per Plot: **2.4mAh per insertion**



Pre-test



Post-test